6.1 Derivative function and standard functions

Definition and results

A function is differentiable on an interval when it is differentiable at each of its points; its derivative associates with x the number $f\prime(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)\prime = nx^{n-1}$ on its domain of differentiability; for $n = 0$, the derivative is 0 directly.

Status and justification. Square: the rate is $2a + h$. Reciprocal: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are accepted here.

Worked example. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f\prime(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists at 0, its standard derivative only for $x > 0$.

Watch out

The domain of differentiability may be smaller than the domain of definition.

Immediate application. Differentiate x^5, x^{-2} and $\sqrt{x}$, stating the domains.

6.2 Sum, product, reciprocal and quotient

Definition and results

For differentiable u and v, $(u + v)\prime = u\prime + v\prime$, $(\lambda u)\prime = \lambda u\prime$, $(uv)\prime = u\prime v + uv\prime$. Where $v \neq 0$, $(\frac{1}{v})\prime = -\frac{v\prime}{v^2}$ and $(\frac{u}{v})\prime = \frac{u\prime v - uv\prime}{v^2}$.

Status and justification. For the product, add and subtract $u(a+h)v(a)$ in the numerator. The rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. Since a differentiable function is continuous (accepted), we obtain $u(a)v\prime(a) + v(a)u\prime(a)$. The other rules are accepted.

Worked example. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f\prime(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping the brackets avoids changing the sign of a single term instead of the whole numerator.

Watch out

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Immediate application. Differentiate $(x^2+1)(3x-2)$, then $\dfrac{1}{x^2+1}$.

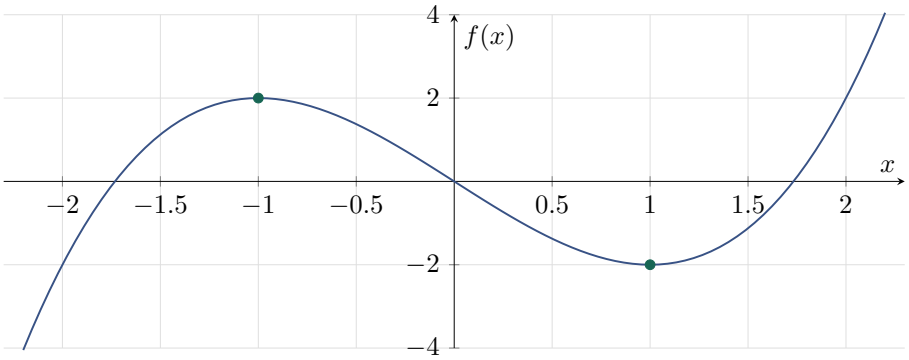
6.3 Sign of the derivative and extrema

Definition and results

On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an interior point where it is differentiable, its derivative is zero there. The converse is false.

Status and justification. Variation theorems accepted. For an interior minimum at a, the rates on the right are positive or zero, those on the left negative or zero; if they converge to the same real number, it is 0. Same argument for a maximum.

Worked example. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $]-\infty;-1]$, decreases on $[-1;1]$, increases on $[1;+\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. Neither of these extrema is global on $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum at -1 and a local minimum at 1.

Watch out

The theorem applies on an interval. A function can decrease on two separate intervals without decreasing on their union.

Immediate application. Study $f(x) = x^2 - 4x + 1$ on $[0;3]$ and give its extrema on this interval.

6.4 Parity, inequalities and comparison

Definition and results

On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The curve of an even function is symmetric about the y-axis; that of an odd function about the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

Status and justification. The symmetries follow from the coordinates of the image points. For an inequality, a study of variation may give a minimum of the difference, then its sign; the derivative does not directly replace this difference.

Worked example. For $x > 0$, $h(x) = x + \dfrac{1}{x} - 2$ has derivative $\dfrac{x^2 - 1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, so $x + \dfrac{1}{x} \geq 2$. Equality is obtained only at 1.

Watch out

Check the domain before parity. The sign of $f\prime - g\prime$ tells us about the variation of $f - g$, not directly about its sign.

Immediate application. Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

6.5 Optimisation and Newton's method

Definition and results

Optimising means defining the variable and its domain, expressing the quantity to optimise, then comparing all the candidate values, including the endpoints. Newton's method uses the tangent to approximate a solution of $f(x) = 0$: $x_{n+1} = x_n - \dfrac{f(x_n)}{f\prime(x_n)}$ when $f\prime(x_n) \neq 0$. It is used only in favourable cases.

Status and justification. Newton's formula comes from the intersection of $y = f(x_n) + f\prime(x_n)(x - x_n)$ with the horizontal axis. Convergence is not guaranteed in general; it is observed and checked in the case chosen.

Worked example. For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton gives $x_1 = \dfrac{3}{2}$, $x_2 = \dfrac{17}{12}$. The squares are 2.25 then about 2.00694: we approach $\sqrt{2}$. The program must check for a zero derivative and a maximum number of steps.

Watch out

A numerical value produced by an algorithm is not a proof of a minimum or of convergence.

Immediate application. A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation.